1 Oct 20-22

1,2,4,6,7

1:$\Delta\nu = \frac{c}{2\mu L} = \frac{3*10^8}{1} = 3*10^8$ Bec : $\Delta\nu = 6*10^8$

So : $n = \frac{\Delta\nu}{3*10^8}$=2.

q could be : q-1;q;q+1.

and $\nu = \frac{qc}{2\mu L}$ so $q = \frac{2\mu L\nu}{c} = \frac{5.85*10^{14}}{3*10^8} = 1.95*10^{16}$and q+1 q-1.

2:

$\Delta\nu = \frac{3*10^8}{2} = 1.5*10^8$=0.1 $*\Delta\nu$

so n=10 ,q could have 11 . if q only neede 1 ,so

$\Delta\nu = 1.5*10^9 = \frac{3*10^8}{2L}$so L=0.1 m .

4:

(1):$P/S = \frac{50W}{(25\mu m)^2\pi}\mu m^2$= K

(2):$\frac{K}{10^4}$=N_1;$\frac{K}{10^3}$=N_2

6:

$\theta = \frac{1.22*632.8nm}{2mm} = \theta_1$

θ_0=$\frac{\lambda}{\pi\omega_0}$=$\frac{\lambda}{\pi\sqrt{\frac{\lambda L}{2\pi}}}$

7:

ω_0=$\sqrt{\frac{\lambda L}{2\pi}}$

$\omega(z)$=$\omega_0\sqrt{1+\frac{\lambda z}{\pi\omega_0^2}^2}$

$L = 0.4; \lambda = 0.6328\mu m; z = 56cm$

9:

$R = z[1+(\frac{f}{z})^2]\frac{3\sqrt{3}}{7} = f, R_1' = R_2' = \frac{6\sqrt{3}}{7}$

$\omega_0 = \sqrt{\frac{\lambda f}{\pi}} = \sqrt{\frac{\lambda 3\sqrt{3}}{7\pi}}$

$\omega = \omega_0\sqrt{1+(\frac{f}{z})^2} = \sqrt{\frac{\lambda 3\sqrt{3}}{7\pi}}\sqrt{1+(\frac{3\sqrt{3}}{7z})^2}$

$\frac{R(z_1)-z_1}{z_1} = (\frac{f}{z_1})^2; \frac{R(z_2)-z_2}{z_2} = (\frac{f}{z_2})^2$

$\frac{\frac{1.5-z_1}{3+z_1}}{} = \frac{1-z_1}{z_1}; z_1+z_2 = 1; \frac{6}{7} = z_1; z_2 = \frac{1}{7};$

10:

$2 = 1[1+f^2]$;f=1;$\omega = \omega_0\sqrt{1+(\frac{f}{z})^2} = \sqrt{\frac{2\lambda}{\pi}} = 2.5977mm$

$\omega = \omega_0\sqrt{1+(\frac{f}{z})^2} = \omega_0\sqrt{1+f^2} = \omega_0\sqrt{R} = \sqrt{\frac{\lambda\sqrt{R-1}R}{\pi}} = 0.3$

$0.3^2\pi/\lambda = R\sqrt{R-1}$;R=41.7733cm.

$\omega_0 = \sqrt{\frac{\lambda\sqrt{R}}{\pi}}$

$\omega = \omega_0\sqrt{1+f^2}; R = 1+f^2; f = \sqrt{R-1}; \omega_0\sqrt{R} = \sqrt{\frac{\lambda R\sqrt{R}}{\pi}}$

$R = (0.3^2\pi/\lambda)^{2/3} =$

$f = \frac{\sqrt{2(R-2)^2(2R-2)}}{2R-4} = \frac{2\sqrt{R-1}}{2} = \sqrt{R-1}$

$\omega_0 = \sqrt{\frac{\lambda\sqrt{Rz}}{\pi}\frac{R}{z}}; f = \sqrt{(\omega_0^2\pi/\lambda)^{2/3}}$

$(\omega_0^2\pi/\lambda)^{2/3} = R$

$\omega = \omega_0\sqrt{1+(\omega_0^2\pi/\lambda)^{2/3}}$